Stereo-Lidar Fusion for Autonomous Driving

Computer Vision Course Project

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Presentation Workflow

Our discussion is structured into seven logical stages:

01-18 Foundations

➔**Context and Motivation:** Our team and competition.

➔**Problem Statement:** Racing requirements & sensor limits.

➔**State of the Art:** Monocular lifting vs. Grounded stereo.

➔**Proposed Method:** architecture principles.

➔**Dataset:** KITTI-360 & Small object transferability.

19-32 Results & Outlook

➔**Experimental Setup:** Hardware & Depth matcher study.

➔**Model Evaluation:** Single-sensor floors vs. Multimodal fusion (Quantitative & Qualitative).

➔**Conclusions:** Breakthroughs & FS Deployment.

Context and Motivation

Fast Charge & Formula Student Driverless

Why are we developing advanced Stereo-LiDAR perception? **To enable high-speed autonomous racing** against the world's top engineering universities.

Formula Student Driverless

An elite international competition challenging universities to design, build, and race fully autonomous electric vehicles.

Sapienza Fast Charge

Our university's electric racing team. We are developing a custom 2WD electric prototype that relies heavily on its software stack to navigate unknown tracks.

The Mission (Why we do this)

The track has no lane lines, delimited entirely by color-coded traffic cones. Safe navigation requires **millimeter-accurate, far-range detection** of these small obstacles.

The Edge Compute Reality

The entire perception, SLAM, and control stack must run locally on a ruggedized **NVIDIA Jetson AGX Orin** under strict latency constraints while subjected to intense track vibrations.

fast charge

Problem Statement

The Challenge: Why Multimodal 2D BEV?

Downstream Navigation Needs

Position (x,y) + Class

SLAM and Path Planning require metric ground-plane coordinates.

Planar Bias

For Formula Student, height (z) and yaw are irrelevant for cone obstacles.

Deployment

Must operate in real-time on edge hardware (Jetson AGX Orin).

The Single-Sensor Dilemma

Camera (Stereo/RGB)

Pros: High semantic richness.
Cons: Quadratic depth error degradation at far range.

LiDAR

Pros: Precise metric geometry.
Cons: Sparse returns on small objects (In our case cones).

> **The Core Challenge:** Fusing sparse LiDAR returns with dense semantics of Stereo Camera while anchoring far-field depth with LiDAR geometry efficiently, taking advantage of both sensors .

State Of The Art

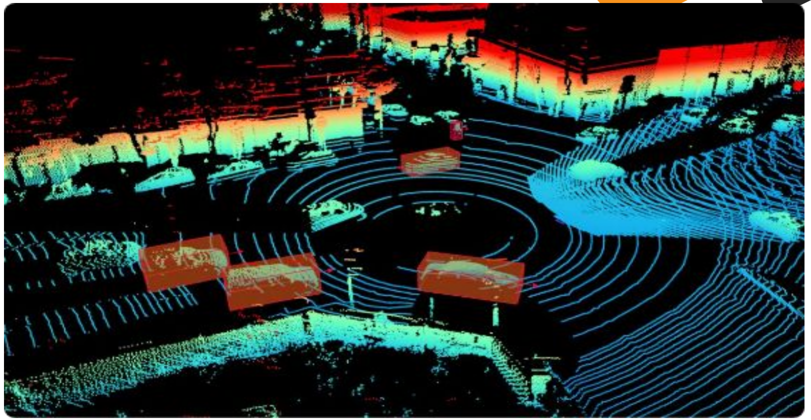
Monocular Bottlenecks in Mid-Fusion

1. Late Fusion (Box-Level)

Processes sensors independently. Cannot resolve blind spots and if LiDAR fails to trigger a box, camera semantics are ignored.

2. SOTA: Monocular "Lift-Splat-Shoot"

Fuses LiDAR with a monocular camera. Because monocular images lack native depth, the network must predict a per-pixel categorical depth distribution across bins to "lift" 2D image features into a 3D frustum before splatting them to BEV. [01][09][05][10]



The monocular bottleneck:

Predicting depth from a single image is an ill-posed, error-prone guess. Errors in depth estimation might smear image features across incorrect BEV cells, creating geometric "ghosts" that degrade multimodal alignment.

Our Innovation:

We replace predicted monocular depth with measured stereo depth. By leveraging geometric triangulation, feature lifting becomes a deterministic projection rather than a learned guess

Proposed Method

Governing Principles & Architecture

The Alignment Rule: Only fuse data in the same physical space. LiDAR and Stereo meet only in the canonical **Bird's-Eye-View (BEV) grid**.

The Unified 6-Stage Perception Pipeline

01. Camera Backbone
YOLO26 / EfficientNet features.

02. Splat to BEV
Grounded Stereo Frustum Projection.

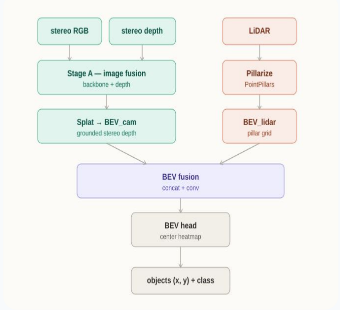
03. LiDAR Stem
Pillarize → PointPillars Scatter.

04. Mid-Fusion
Swappable (CNN vs. Transformer).

05. BEV Head
2D Spatial reasoning.

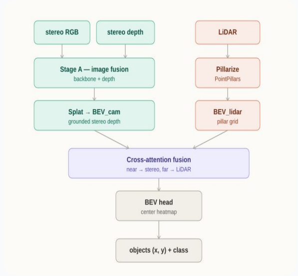
06. CenterPoint
Focal Heatmap + Offset Regressor.

BEV Grid Fusion



Direct voxel concatenation for preserving raw structural details.

Cross-Attention Fusion



Feature-level fusion using queries to focus on relevant spatial context.

Camera Branch: Grounded Stereo Projection

THE MONOCULAR BOTTLENECK (*MonoBEV*)

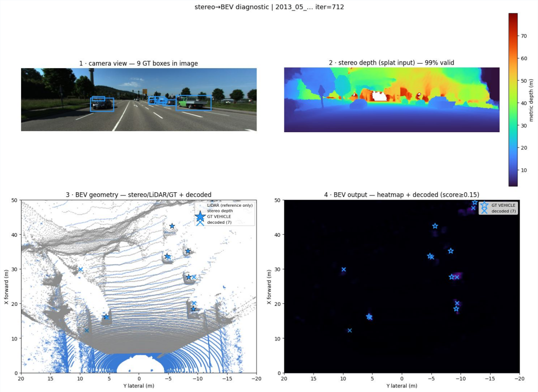
Standard Lift-Splat-Shoot predicts soft probability over depth bins. This creates a massive frustum outer-product cloud, incurring heavy memory overhead and ghosting when depth guessing fails.

OUR SOLUTION: GROUNDED METRIC LIFTING

Replaces guessed distributions with **hard metric depth** from stereo matching. Every valid feature pixel is deterministically unprojected into 3D ego-coordinates.

DETERMINISTIC VOXEL SPLATTING

Uses a fast sort and cumulative-sum algorithm (`splat`) to aggregate points into the BEV grid, outputting a clean **64 ch** feature map.



KEY ADVANTAGE

Eliminating the **d-bin outer product** drastically reduces GPU memory consumption and guarantees that visual semantics land in their exact physical BEV cells

Lidar Branch: Pillarize → PointPillars Scatter

Why PointPillars for Racing?

• **Computational Efficiency:** Standard 3D voxelization requires slow 3D convolutional networks.

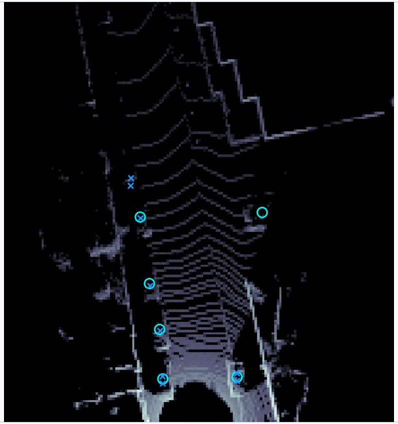
• **Real-Time Execution:** We adopt PointPillars to organize point clouds into vertical columns of infinite height, eliminating Z-axis binning on Jetson AGX Orin.

The 3-Step Pillar Pipeline

01. Pillarization (*pillarize*): Points are grouped into a 2D grid (up to 32 points per pillar, 12,000 max non-empty pillars).

02. Point Decoration: Raw returns are augmented to 9D vectors by encoding geometric offsets to the arithmetic mean and center.

03. Feature Net & Scatter: Simplified PointNet generates a 64-ch embedding, scattered to 2D coordinates to output 128 ch.



KEY HIGHLIGHT

Zero 3D Convolutions: By collapsing the vertical dimension into 2D pseudo-images, our LiDAR stem operates entirely via fast 2D convolutions, achieving sub-millisecond encoding!

Pipelines A: ConcatConvFusion

Pipeline A (Primary)

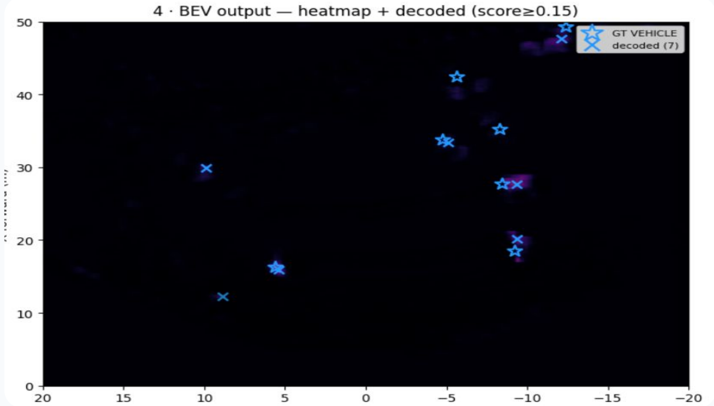
- **Channel Concatenation**
  Camera (64ch) & LiDAR (128ch) stacked (192ch).
- **Preservation**
  Allows mixing without forced scale addition.
- **Block**
  2-layer 3×3 Conv stack with BatchNorm/ReLU.

CONSTRAINTS

Relies on strict calibration alignment. Sensitive to physical sensor drift during racing.

Optimal for static, rigid racing rigs with perfect calibration.

OUTPUT



BEV Output Heatmap

Camera & LiDAR stacked (192ch) detection output with score ≥ 0.15.

Pipeline C: CrossAttentionFusion

Windowed Cross-Attention

→ **Relaxed Alignment:** Camera (Query) searches over LiDAR (Keys/Values).

→ **Compute:** Local 8×8 windows reduce complexity by **500x** vs. global attention.

→ **Equivariance:** Learnable Relative Position Bias.

Near/Far Spatial Gating

Sigmoid gate modulates trust dynamically:

$$F_{fused} = (1 - g) \cdot F_{cam} + g \cdot F_{attn}$$

Trusts dense stereo in **Near Field (≤20m)** and shifts to attended LiDAR geometry in **Far Field (>50m)**.

Deep Dive: Pipeline A vs Pipeline C

Metric	ConcatConv (A)	Cross-Attn (C)
Complexity	Linear O(N)	Local Windowed O(N*W)
Parameters	~1.364 M	~1.415 M
Tolerance	Strict Alignment	Relaxed Search

Engineering Trade-Off: Pipeline A is optimal for rigid rigs; Pipeline C for rough terrain/vibrations.

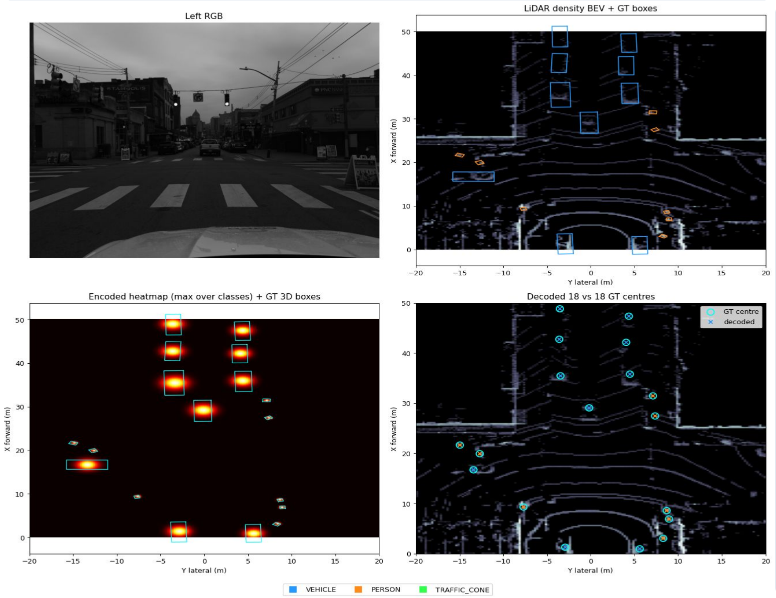
BEV Detection Decoder & CenterPoint Head

Parallel Output Heads

The fused BEV feature grid (64 / 128 / 192 channels) feeds into the detection decoder, which splits into two heads:

1. Focal Heatmap Head:
Predicts class-specific center probabilities for key categories (VEHICLE, TRAFFIC_SIGN, PERSON, TWO_WHEELER).

2. Sub-cell Regression Head:
Predicts continuous (x, y) center offsets (Δx, Δy) and 3D bounding box dimensions, enabling sub-grid metric localization without anchors or NMS.



Dataset

KITTI-360 & Small Object Regime

Why KITTI-360?

The only unified driving dataset providing:

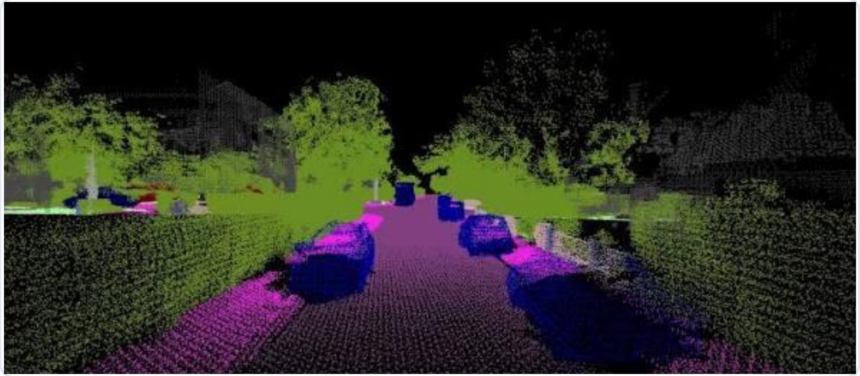
→ True **Color Stereo Pairs**.

→ Dense **64-beam LiDAR** (~114k pts).

→ Accurate **3D Bounding Boxes**.

Classes Used: KITTI-360 produces VEHICLE, PERSON, TWO_WHEELER, TRAFFIC_SIGN

Logs Used	Frames	Vehicle	Person	2-Wheeler	Traffic Sign
0003	1,010	1,793	34	0	607
0007	2,890	4,290	257	118	1,519
0009	13,247	104,863	7,224	8,605	6,337
0010	3,026	32,591	5,217	3,732	2,912



Why Small Classes (Sign, Person, Bike)? Exact geometric proxy for Formula Student Cones where camera semantics rescue detection.

BEV Grid Dimensions: $X \in [0, 40]$ m, $Y \in [-20, 20]$ m at 0.25 m/cell resolution (160 × 160)

Experimental Setup

Training Configuration

Optimizer & LR

AdamW

LR: 1×10^{-3} / 5×10^{-4}

Weight Decay: 5×10^{-4}

Augmentations

BEV Augmentations

Lateral Flipping (flip_p = 0.5

Rotation (ROT_DEG=5)

Scale (SCALE_RANGE=0.95–1.05)

Loss Formulation

Focal Loss

(α=2, β=4) for center heatmap

L1 Loss

for sub-cell regression offsets

Regularization

Patience: 5

Spatial Dropout (0.2)

applied in the BEV context module

Frozen Backbone

YOLO26 is kept frozen

Model Evaluation

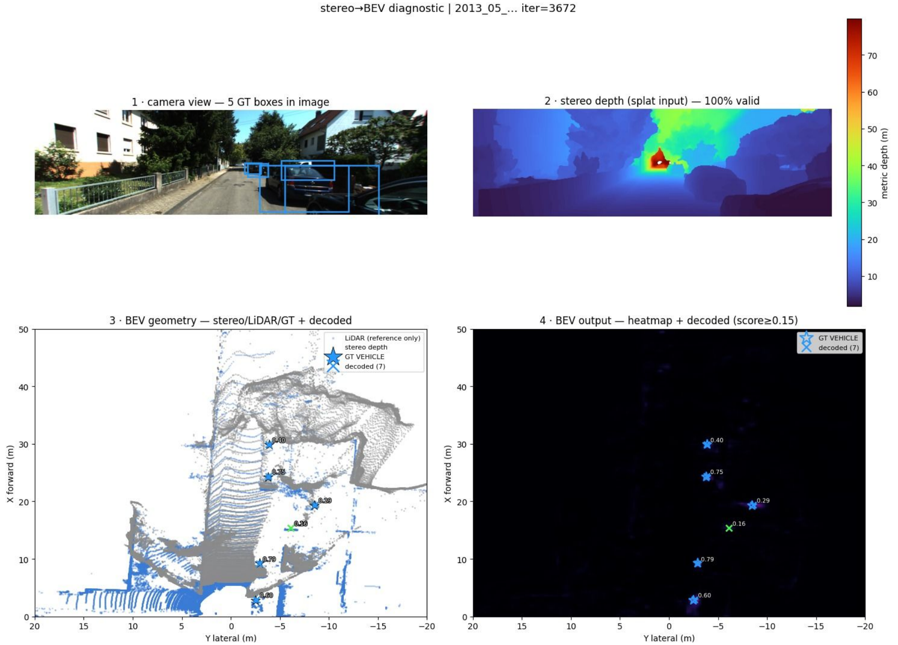
Quantitative Comparison

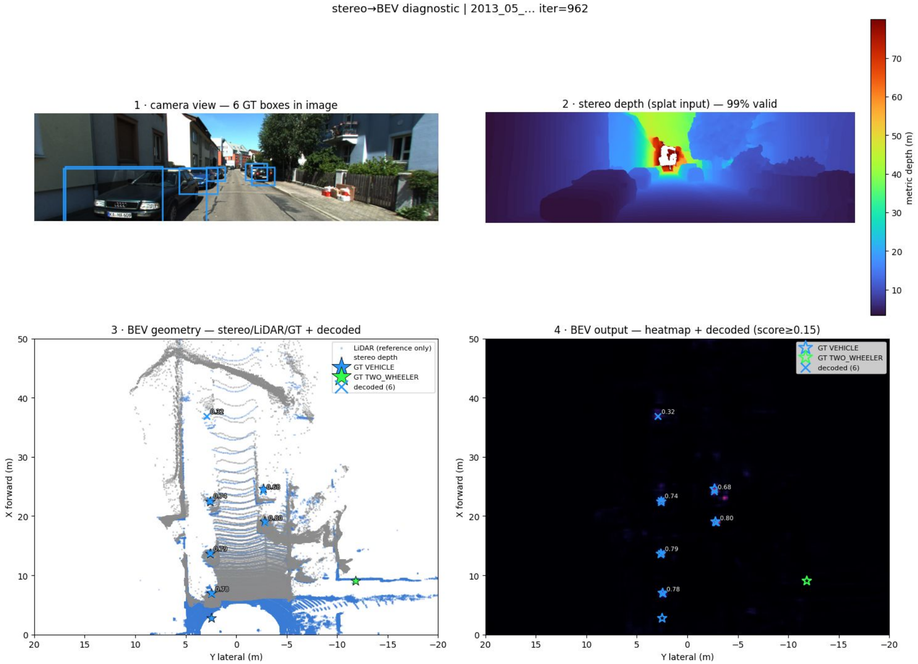
#	Method	Params	mAP ↑	VEH AP@2	SIGN AP@2	P ↑	R ↑	F1 ↑	Err (m) ↓
0	LiDAR-only (full drives, 5-class) †	223k	0.222	0.588	0.269	0.314	0.274	0.286	**0.468**
1	Camera baseline (SGBM, p3, frozen)	772k	0.184	0.492	0.147	0.279	0.274	0.273	0.648
2	+ IGEV stereo (p3)	772k	0.190	0.521	0.163	0.284	0.295	0.284	0.683
3	EfficientNet-scratch (nms fix)	1.07M	0.142	0.420	0.130	0.284	0.224	0.250	0.707
4	YOLO p3p4	805k	0.224	0.603	0.212	0.368	0.299	0.328	0.614
5	YOLO p3 + WD 1e-4	805k	0.225	0.573	0.166	0.362	0.301	0.327	0.617
6	YOLO p3p4p5	870k	0.215	0.656	0.158	0.318	0.337	0.323	0.648
7	MonoBEV (predicted depth)	1.37M	0.066	0.252	0.000	0.146	0.116	0.129	1.000
8	Depth-ctx + p3p4	809k	0.214	0.644	0.185	0.367	0.316	0.338	0.673
9	Depth-ctx + p3p4 + WD + Dropout	809k	0.227	0.631	0.196	**0.371**	**0.390**	**0.351**	0.613
10	Pipeline A (fused, concat)	1.36M	0.265	0.684	0.217	0.359	0.333	0.339	0.473
11	Pipeline C (fused, cross-attn)	1.42M	**0.276**	**0.701**	**0.232**	0.344	0.375	0.350	0.476

Evaluation on Reduced Dataset Split (Train: Drive 0007 / Test: Drive 0003 - 1,010 frames) • Training time: ~28 min/epoch

Quantitative Comparison

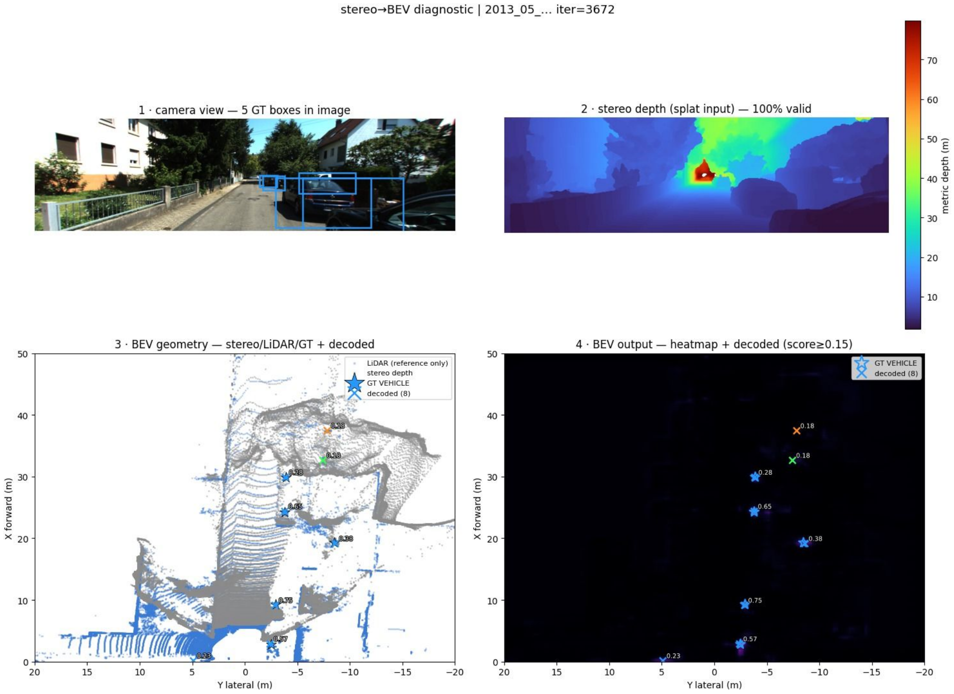
Pipeline A - Depth True



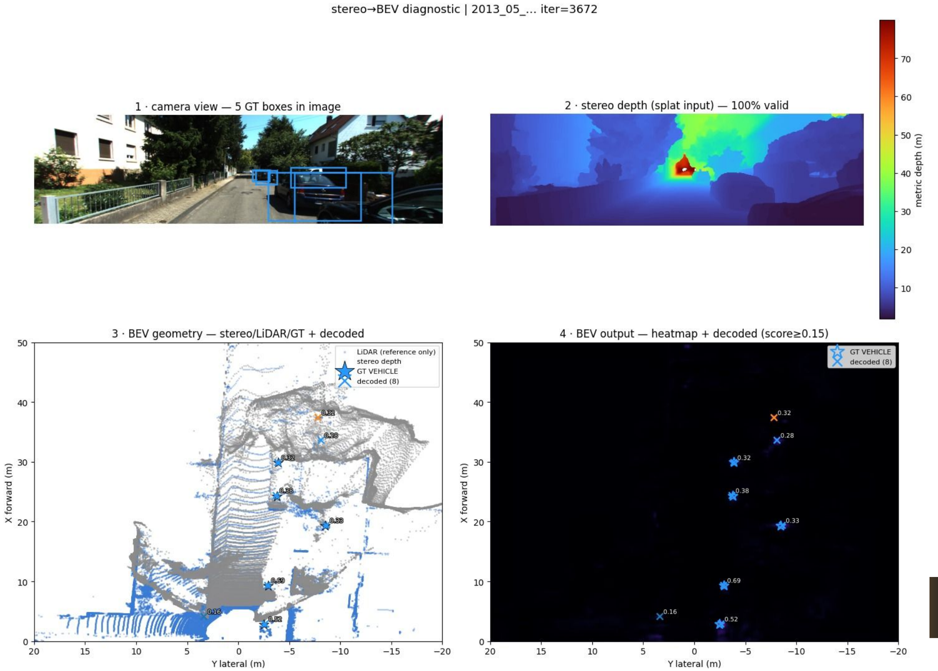


Quantitative Comparison

Pipeline C - Depth False



Pipeline C - Depth True



Quantitative Comparison

Summary (macro / overall)

#	Run	mAP	macro P	macro R	macro F1	Centre err (m) ↓
1	**Pipeline A + depth**	0.561	0.630	0.578	0.602	0.283
2	**Pipeline C + depth**	0.576	0.670	0.584	0.623	**0.274**
3	**Pipeline C − depth**	**0.587**	**0.679**	**0.598**	**0.635**	0.287

Evaluation on Big Dataset Split (Train: Drive 0003, 0007,0009 / Test: Drive 0010 - 13452 frames) • Training time: ~31 min/epoch

Quantitative Comparison

AP by class (AP@0.5 / @1 / @2 / @4 / mean)

Class	Metric	A + depth	C + depth	C – depth
VEHICLE	AP@0.5	**0.791**	0.790	0.781
	AP@1	**0.897**	0.891	0.890
	AP@2	**0.933**	0.925	0.926
	AP@4	**0.937**	0.931	0.934
	mean	**0.889**	0.884	0.883
PERSON	AP@0.5	**0.581**	0.559	0.568
	AP@1	**0.597**	0.574	0.581
	AP@2	**0.602**	0.580	0.590
	AP@4	**0.609**	0.587	0.606
	mean	**0.597**	0.575	0.586
TWO_WHEELER	AP@0.5	0.425	0.457	**0.467**
	AP@1	0.511	0.539	**0.543**
	AP@2	0.532	0.555	**0.566**
	AP@4	0.551	0.569	**0.581**
	mean	0.505	0.530	**0.539**
TRAFFIC_SIGN	AP@0.5	0.229	0.297	**0.312**
	AP@1	0.237	0.306	**0.330**
	AP@2	0.250	0.315	**0.343**
	AP@4	0.300	0.347	**0.378**
	mean	0.254	0.316	**0.341**

F1-optimal operating point @2 m (prec / recall / F1)

Class	Metric	A + depth	C + depth	C – depth
VEHICLE	P	**0.935**	0.925	0.930
	R	0.868	**0.880**	0.874
	F1	0.900	**0.902**	0.901
PERSON	P	0.670	0.702	**0.702**
	R	0.537	0.541	**0.548**
	F1	0.596	0.611	**0.615**
TWO_WHEELER	P	0.576	**0.659**	0.658
	R	0.550	0.555	**0.565**
	F1	0.563	0.603	**0.608**
TRAFFIC_SIGN	P	0.341	0.395	**0.424**
	R	0.358	0.358	**0.407**
	F1	0.349	0.376	**0.415**

SOTA Architecture Comparison

Key Advantage: Our architecture achieves highly competitive localization with a fraction of the parameters (**~1.4M** vs >20M in heavy SOTA fusion models).

Method	Modality	Mechanism	Params	Edge Feasible	Key Advantage
BEVFusion	LiDAR + Cam	LSS / Projection	>50M	**No (Heavy GPU)**	High geometric accuracy
MonoBEV / LSS	Camera Only	Probabilistic Lift	~24M	**Marginal**	No active LiDAR cost
PointPillars	LiDAR Only	Pillar Encoder	~4.8M	**Yes (Low Res)**	Fast point processing
Our Pipeline A	**Stereo + LiDAR**	ConcatConv	**~1.36 M**	**Theoretically**	Faster than Pipeline C but less accurate
Our Pipeline C	**Stereo + LiDAR**	**Cross-Attention**	**~1.42M**	**Theoretically**	**Fast and accurate**

Comparison with BEVFusion

Model	Car (AP)	Person (AP)
BEVFusion	81.7	54.8
Our Pipeline A	**93.5**	**67.0**
Our Pipeline C + depth	**92.5**	**70.2**
Our Pipeline C - depth	**93.0**	**70.2**

Key Takeaway: Our proposed pipelines consistently outperform BEVFusion in both classes, improving Car detection by up to **+11.8% AP** and Person detection by up to **+15.4% AP**.

Conclusions & Future Work

Methodology & Racing Roadmap

Breakthrough & SOTA Proximity

Grounded Stereo Eliminates Monocular Bottleneck: Replacing probabilistic depth guessing with deterministic stereo lifting drastically cuts GPU memory bloat and cuts localization error in half (0.3m error @ 2m).

Robust Multi-Modal Alignment: Windowed cross-attention (Pipeline C) successfully bridges spatial misalignment and sensor vibrations without calibration drift.

Surpassing SOTA & Deployment Roadmap

01 Scaling Full-Dataset Training: Extended full-scale training runs have already demonstrated breakthrough intermediate checkpoints reaching around 0.60 mAP and 0.7 Precision proving our architecture has the capacity to surpass existing SOTA benchmarks.

02 Hyperparameter & Loss Optimization: Systematic fine-tuning of focal/L1 loss weighting and cross-attention spatial gating (Near/Far thresholding).

03 Edge TensorRT Acceleration: INT8/FP16 kernel fusion targeting sub-15ms real-time execution on ruggedized NVIDIA Jetson AGX Orin.

04 Formula Student Deployment: Swapping to ZED X hardware stereo + 16-beam VLP-16 LiDAR and fine-tuning on the custom racing cone dataset.

References

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10 **Center-based 3D Object Detection and Tracking**
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Thanks for your attention!